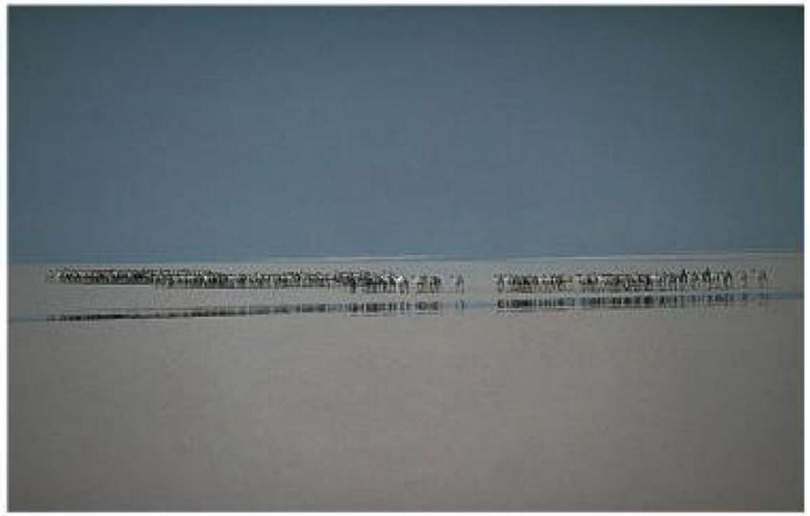
PHYS 104 – PHYSICS II

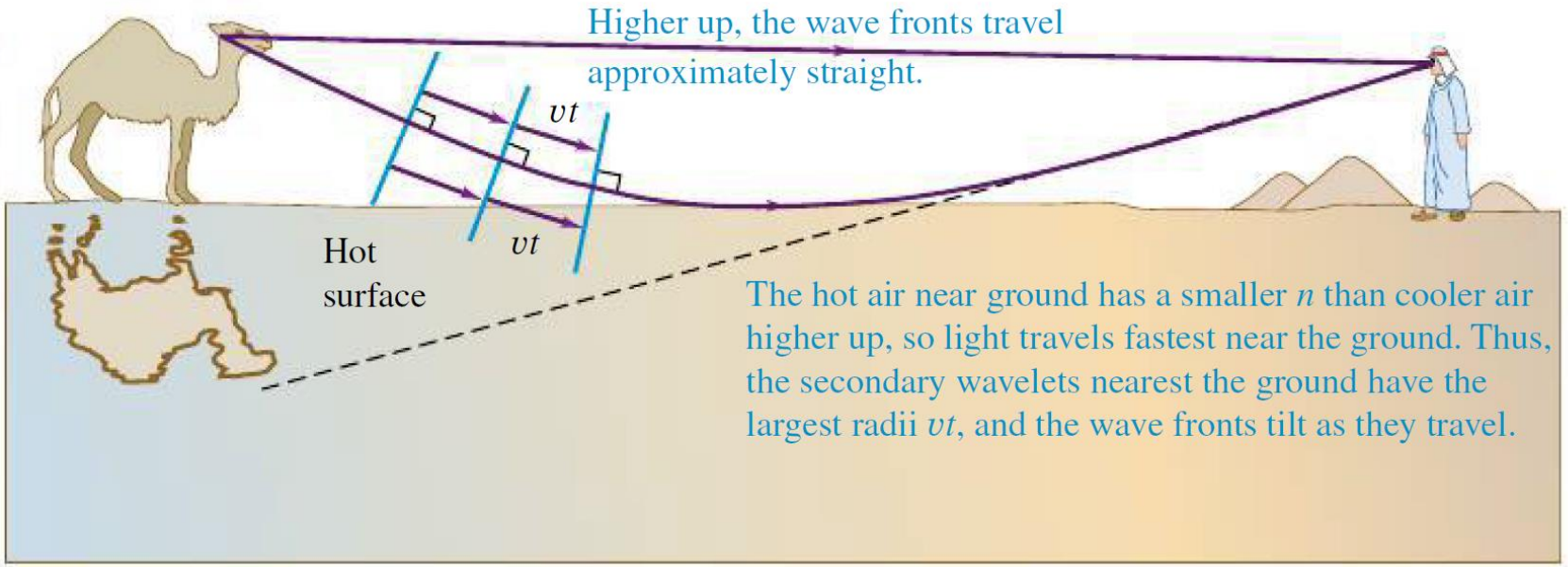
DR. ASHRAF MOUSA ABDELWAHED

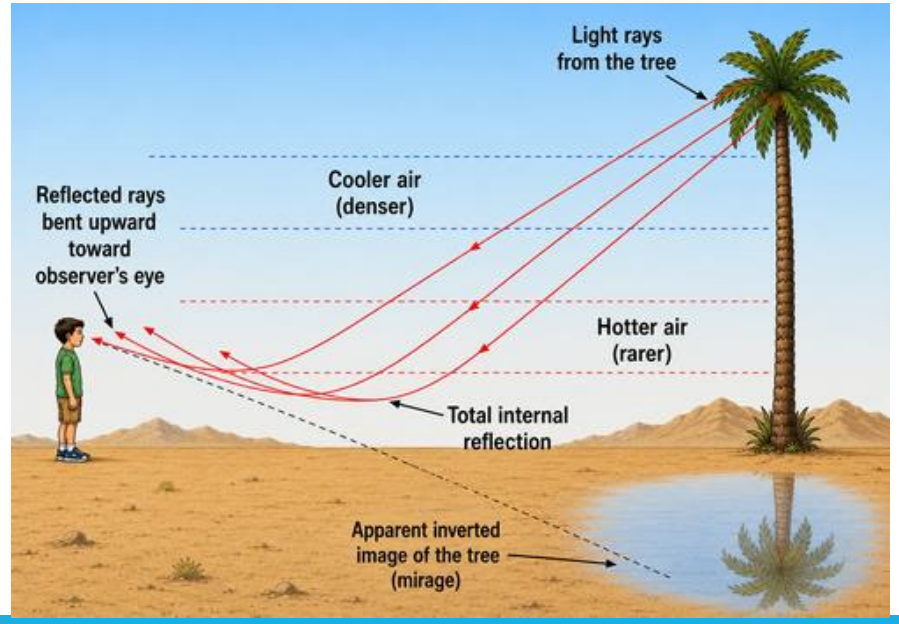
FALL 2026

Lecture 3

Mirage



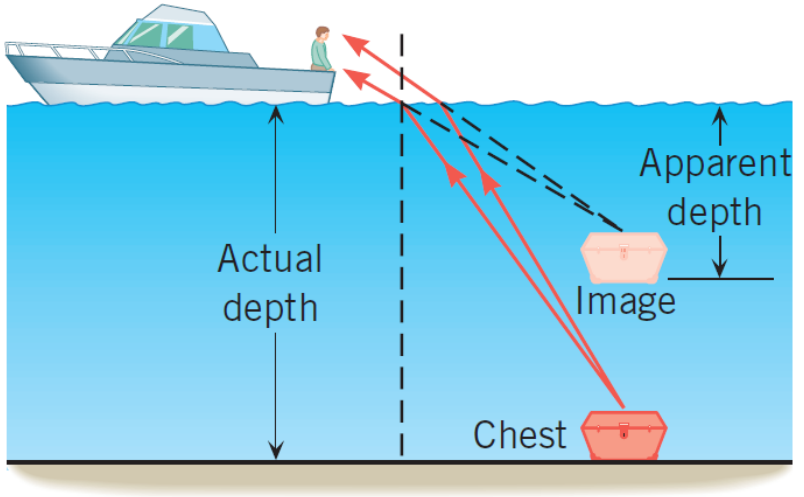

Higher up, the wave fronts travel approximately straight.
vt
vt
Hot surface
The hot air near ground has a smaller n than cooler air higher up, so light travels fastest near the ground. Thus, the secondary wavelets nearest the ground have the largest radii vt, and the wave fronts tilt as they travel.


Light rays from the tree
Reflected rays bent upward toward observer's eye
Cooler air (denser)
Hotter air (rarer)
Total internal reflection
Apparent inverted image of the tree (mirage)

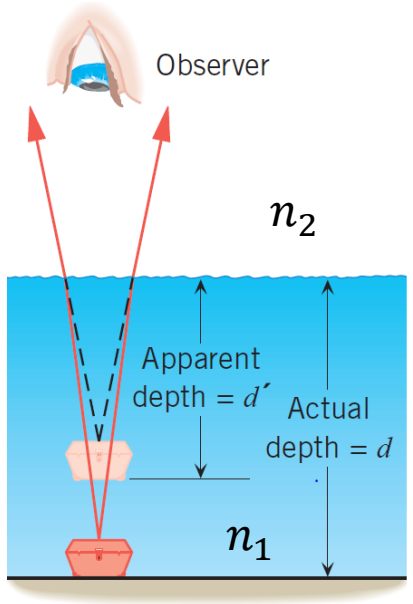
Apparent Depth

- An object lying under water appears to be closer to the surface than it actually is.
- For the situation shown in Figure “a”, it is difficult to determine the apparent depth.
- A much simpler case is shown in part “b”, where the observer is directly above the submerged object, and the apparent depth d' is related to the actual depth d by:

$$d' = d\frac{n_2}{n_1}$$



(a)



(b)

Example 4:

A swimmer is treading water (with her head above the water) at the surface of a pool 3.00 m deep. She sees a coin on the bottom directly below. How deep does the coin appear to be?

Solution

$$d' = d\frac{n_2}{n_1} = 3 \times \frac{1}{1.33} = 2.26\ \text{m}$$

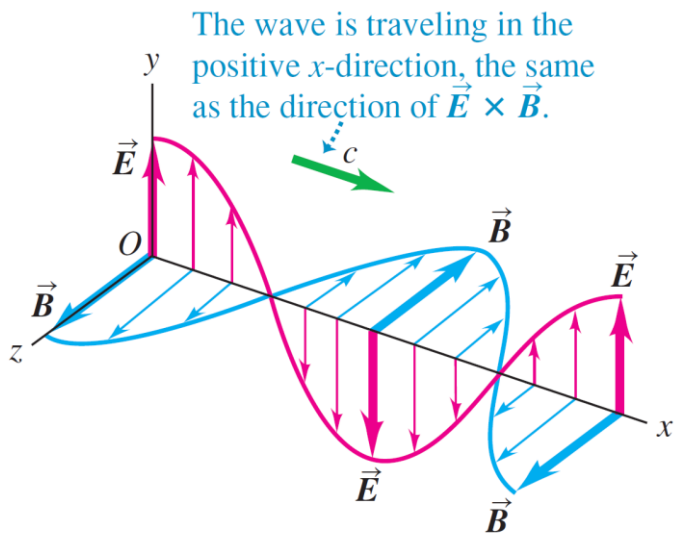
Polarization of light

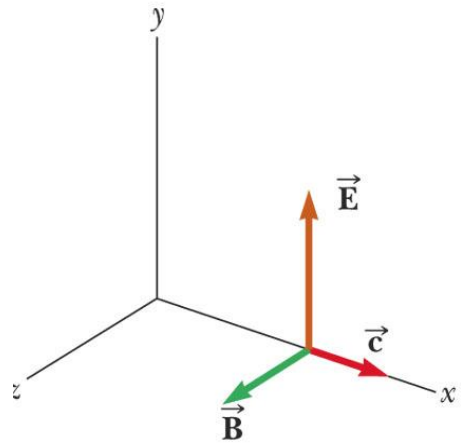
Electromagnetic Waves "EMW.":

- It consists of an electric field & a magnetic field perpendicular to each other & both perpendicular to direction of propagation.
- Any accelerating charge (electron "e" or proton "p") emits EMW
- EMW is transverse Wave & light is EMW
- EMW travels with constant speed in vacuum "space" given by:

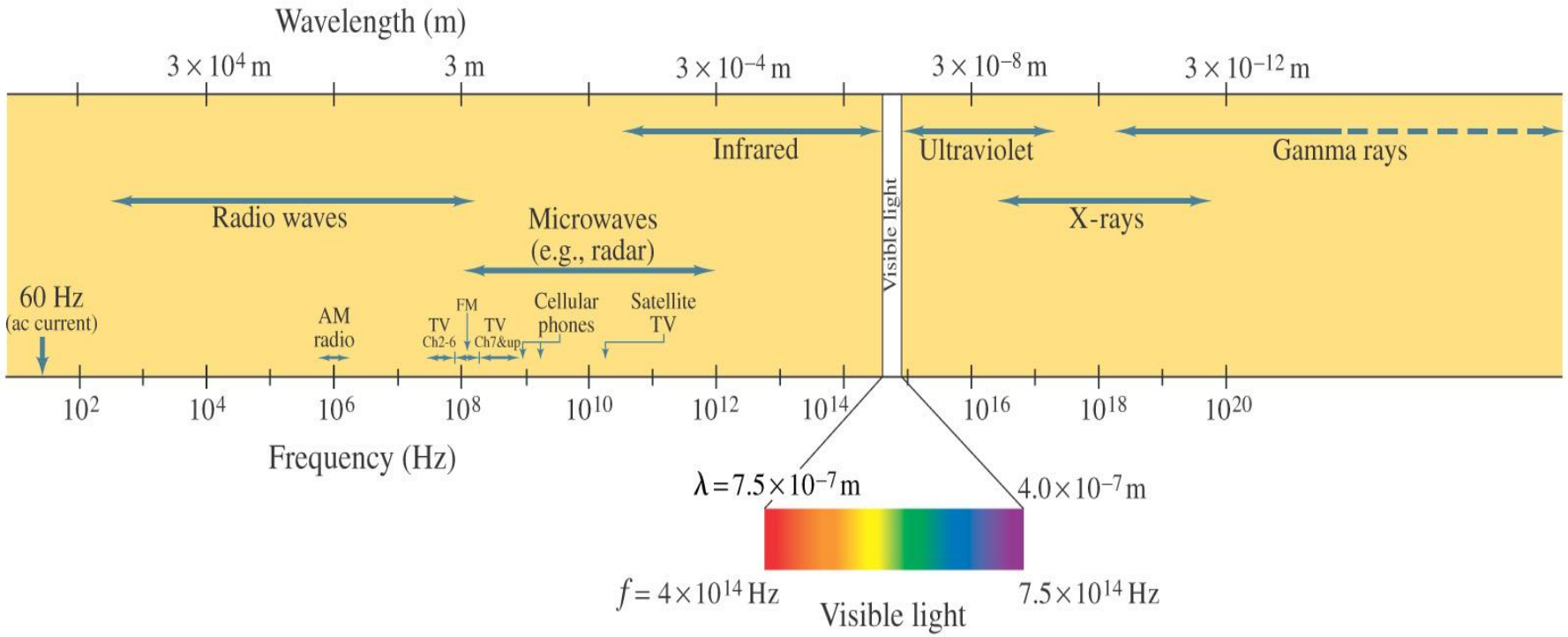
$$\boldsymbol{c = 3 \times 10^{8}\ m/sec}$$

- The velocity v of E.M.W in any medium is less than its velocity in air c.





Electromagnetic spectrum:



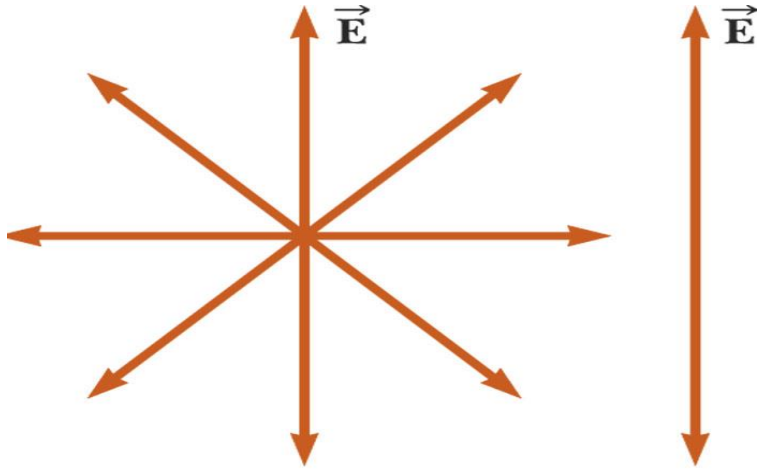
As λ decreases, f increases & Energy increases

As λ increases, f decreases & Energy increases

Polarization of light

Linear Polarization of EMW:

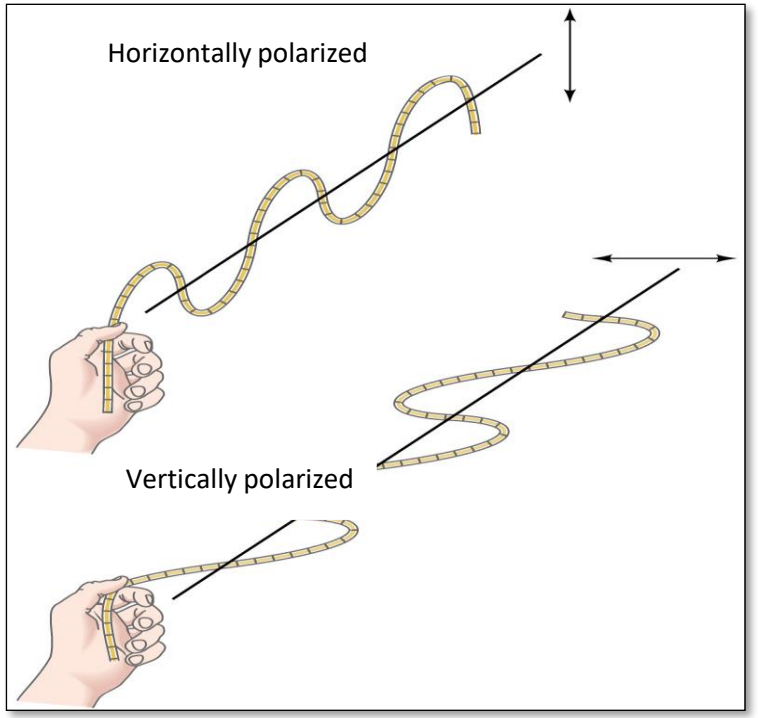
- A linearly polarized E.M.W. is the wave in which the electric field vibrates in one direction only instead of all directions



Unpolarized light polarized light

Explanation of polarization:

- Consider a rope is fixed from one end & the other end is vibrated in vertical direction. The wave produced in the wave is vibrating only in vertical direction; therefore it is polarized in vertical direction.
- If the rope is vibrated in horizontal direction. The wave produced in the wave is vibrating only in horizontal direction; therefore it is polarized in horizontal direction.



(a) A linearly polarized light wave can pass easily through a slit parallel to its direction of polarization

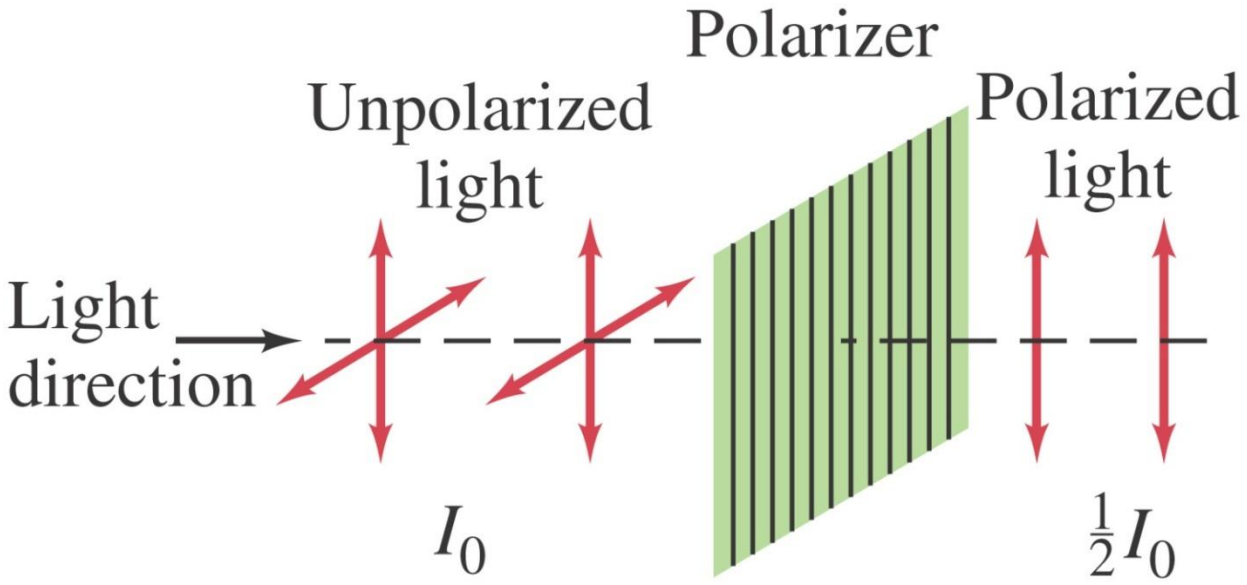
(b) A linearly polarized light wave can pass easily through a slit perpendicular to its direction of polarization

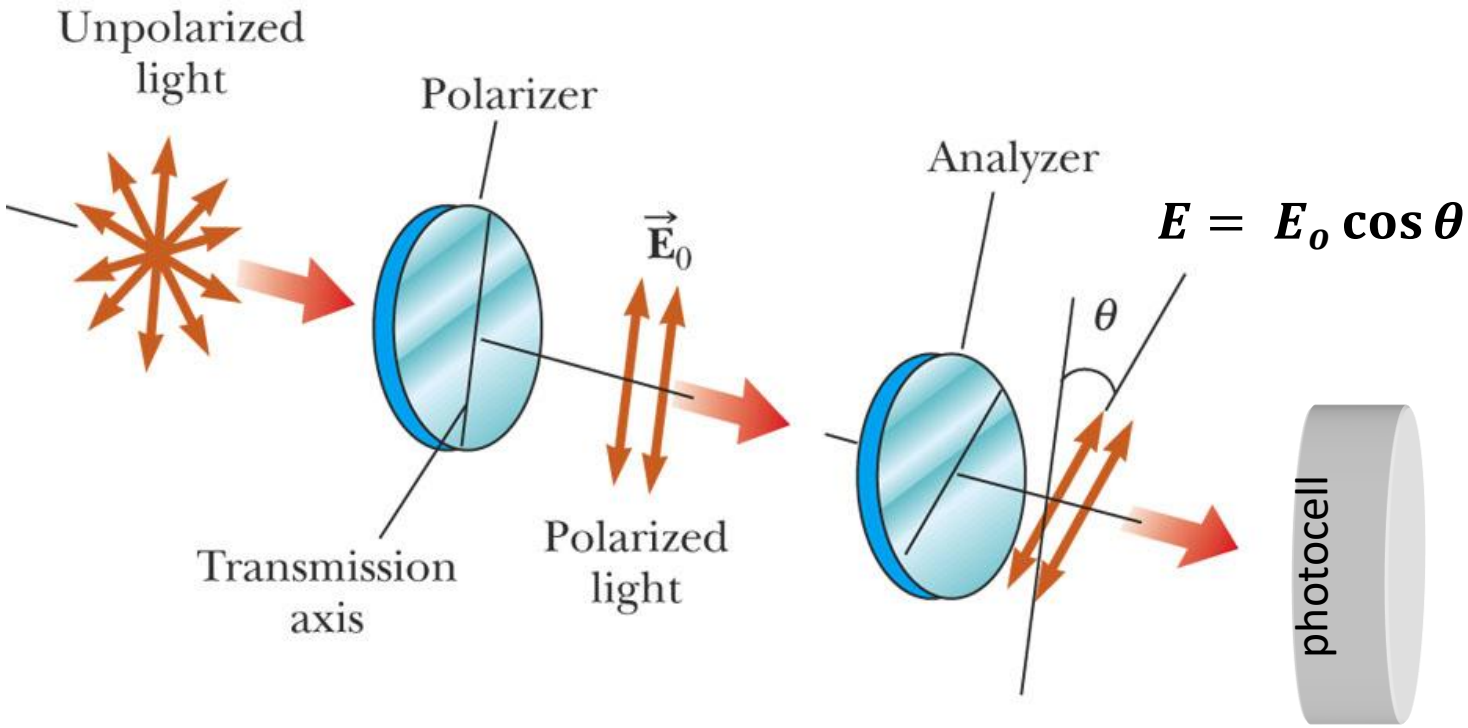


Methods of polarization:

I- Polarization by selective absorption:

- This is done by a Polaroid which is a crystal with a certain transmission axis, it allow the component of electric field parallel to this axis to pass & absorbs the component perpendicular to this axis.
- Intensity of transmitted polarized light is half the intensity of incident unpolarized light.





θ: angle between the axis of polarizer & the axis of analyzer

E_o: amplitude of electric field of 1st polarized light

E: amplitude of electric field of 2nd polarized light

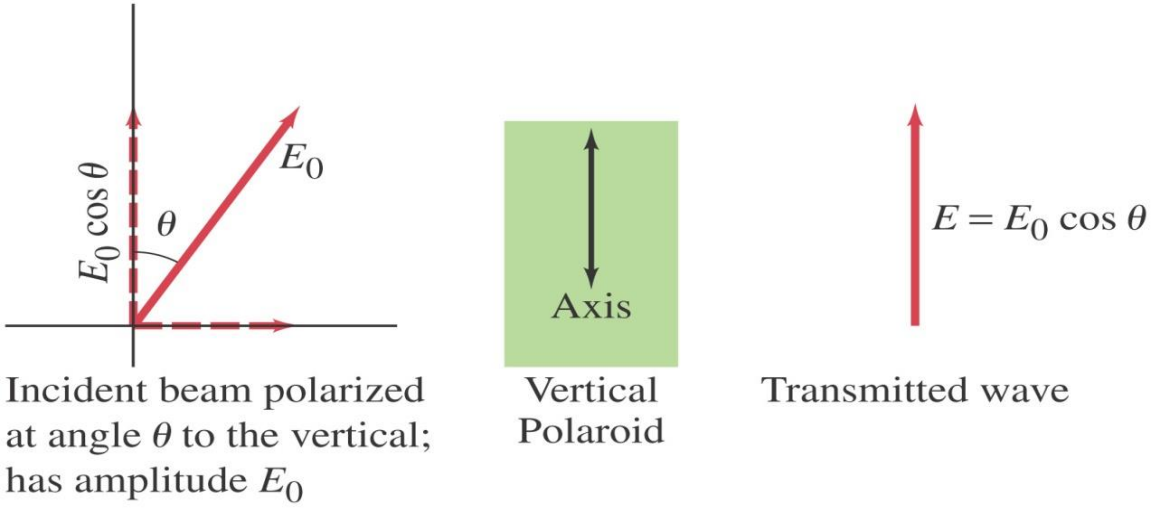
I_o: intensity of 1st polarized light

I: intensity of 2nd polarized light

- E_o is resolved into two components:

1. $\boldsymbol{E_o \cos\theta}$ parallel to the axis of analyzer so it will be transmitted

2. $\boldsymbol{E_o \sin\theta}$ perpendicular to the axis of the analyzer so it will be absorbed



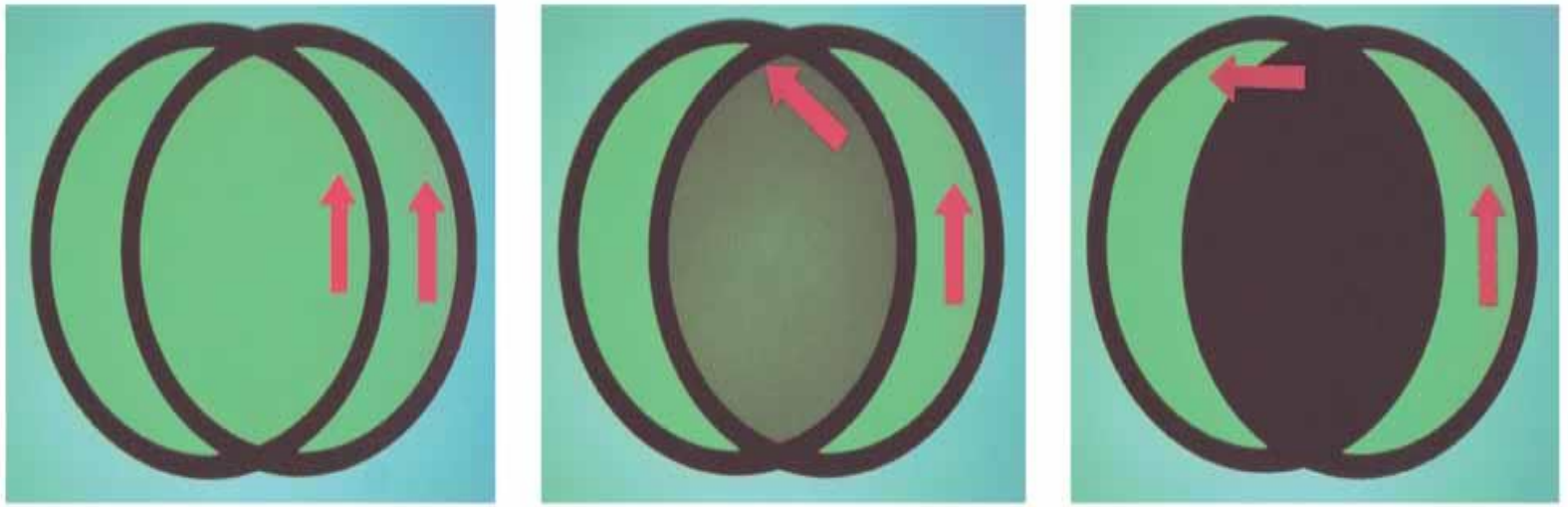
- Since the intensity of light is proportional to square the amplitude of electric field, therefore

$$I_o \propto (E_o)^2 \quad \& \quad I \propto (E_o \cos\theta)^2$$

$$\therefore \frac{I}{I_o} = \left(\frac{E_o \cos\theta}{E_o}\right)^2$$

$$\therefore \boldsymbol{I = I_o \cos^2\theta}$$ **Malus's law**

1. If the polarizer and the analyzer are parallel $(\boldsymbol{\theta = 0°})$ then $\boldsymbol{I = I_o}$(maximum)
2. If $\boldsymbol{\theta = 45°}$ then $\boldsymbol{I = {I_0}/{2}}$
3. If the polarizer and the analyzer are perpendicular $(\boldsymbol{\theta = 90°})$ then $\boldsymbol{I = 0}$ (minimum)



4. As θ increases from $0°$ to $90°$, the intensity of the final transmitted light decreases.

Example 5:

If the angle between a polarizer & an analyzer used for reducing the intensity of light is 45°. What is the average intensity of the polarized light reaching the photocell if the average intensity of unpolarized light is 1 W/m2.

Solution

$$I_2 = I_1 \cos^2\theta = \frac{I_o}{2}\cos^2\theta = \frac{1}{2}\cos^2 45 = 0.25\, W/m^2$$

- Example 6:

What value of θ should be used so the average intensity of the polarized light reaching the photocell after passing through polarizer-analyzer setup is one-tenth the average intensity of the unpolarized light?

Solution

$$I_2 = I_1 \cos^2\theta = \frac{I_o}{2}\cos^2\theta \quad \& \quad I_2 = \frac{I_o}{10}$$

$$\therefore \frac{I_o}{10} = \frac{I_o}{2}\cos^2\theta \rightarrow \therefore \cos^2\theta = \frac{1}{5}$$

$$\cos\theta = \sqrt{1/5} \longrightarrow \theta = \cos^{-1}\sqrt{1/5} = 63.4°$$

Example 7:

Unpolarized light is incident upon three polarizers. The first polarizer has a vertical transmission axis, the second has a transmission axis rotated 30.0° with respect to the first, and the third has a transmission axis rotated 75.0° relative to the first. If the initial light intensity of the beam is I_b, calculate the light intensity after the beam passes through

a) the second polarizer and

b) the third polarizer.

Solution

(a) Calculate the intensity of the beam after it passes through the second polarizer.

The incident intensity is $I_b/2$. Apply Malus's law to the second polarizer:

$$I_2 = I_1 \cos^2\theta = \frac{I_o}{2}\cos^2\theta = \frac{I_b}{2}\cos^2 30 = \frac{3}{8}I_b$$

(b) Calculate the intensity of the beam after it passes through the third polarizer.

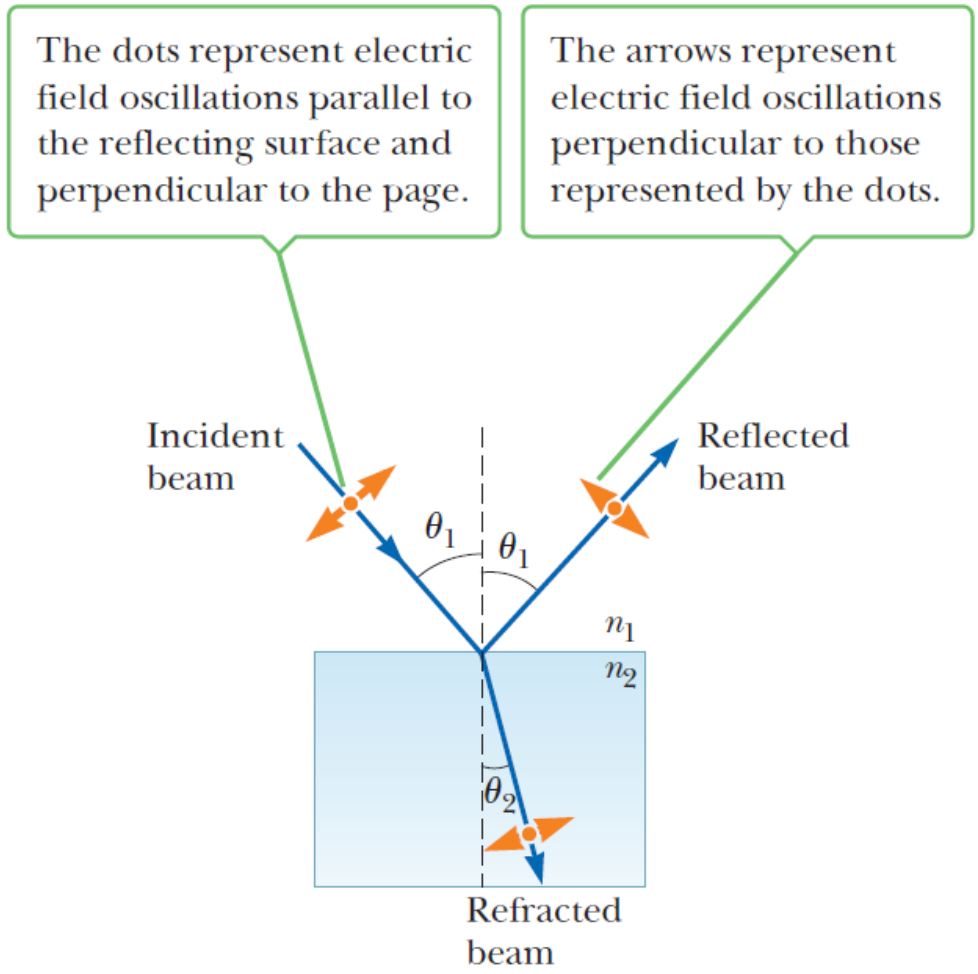
The incident intensity is now $3I_b/8$. Apply Malus's law to the third polarizer.

$$I_3 = I_2 \cos^2\theta = \frac{3}{8}I_b \cos^2 45 = \frac{I_b}{2}\cos^2 30 = \frac{3}{16}I_b$$

Methods of polarization:

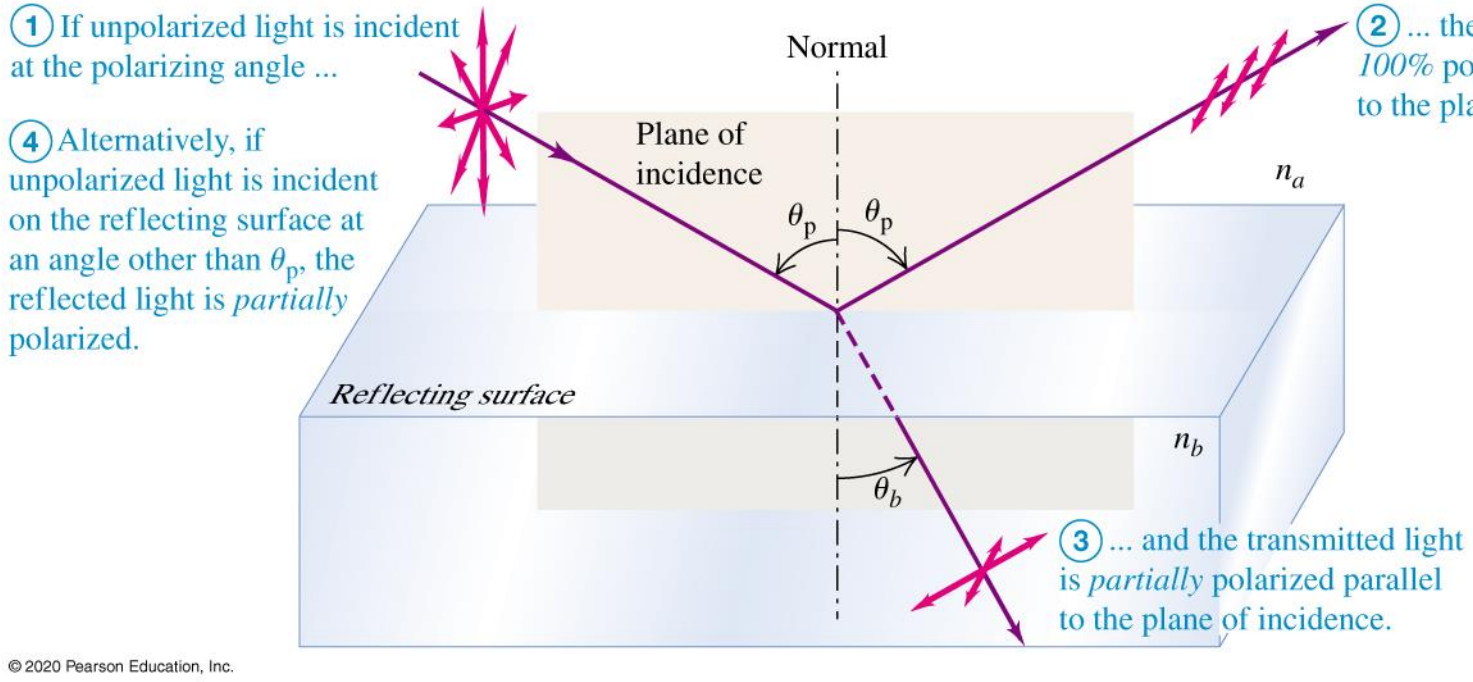
II- Polarization by reflection:

- When unpolarized light is incident from air on a reflecting surface, the reflected beam may be, partially or completely polarized depending on the angle of incidence θ.
- The dots represent the vibrations perpendicular to paper & the arrows represent the vibrations parallel to the paper
- If θ is small then the reflected & refracted rays will be partially polarized.



Methods of polarization:

- At certain angle of incidence θ_P called the polarizing angle, the reflected rays will be completely polarized & the refracted rays are partially polarized. In this case the angle between the reflected and refracted rays is $90°$.



© 2020 Pearson Education, Inc.

$$Snell's\ law: \quad n_1 \sin \boldsymbol{\theta_p} = n_2 \sin(\boldsymbol{\theta_r}) \longrightarrow n_1 \sin \boldsymbol{\theta_p} = n_2 \sin(\mathbf{90} - \boldsymbol{\theta_p})$$

$$n_1 \sin \boldsymbol{\theta_p} = n_2 \cos \boldsymbol{\theta_p},$$

$$\boxed{\tan \boldsymbol{\theta_p} = \frac{\boldsymbol{n_2}}{\boldsymbol{n_1}}} \quad Brewester\ law$$

n_1: *refractive index of first material,* n_2: *refractive index of second material*

For air $n_1 = 1 \rightarrow$ $\boxed{\tan \boldsymbol{\theta_p} = \boldsymbol{n}}$

n: refractive index of reflecting surface θ_p: polarizing "Brewster" angle